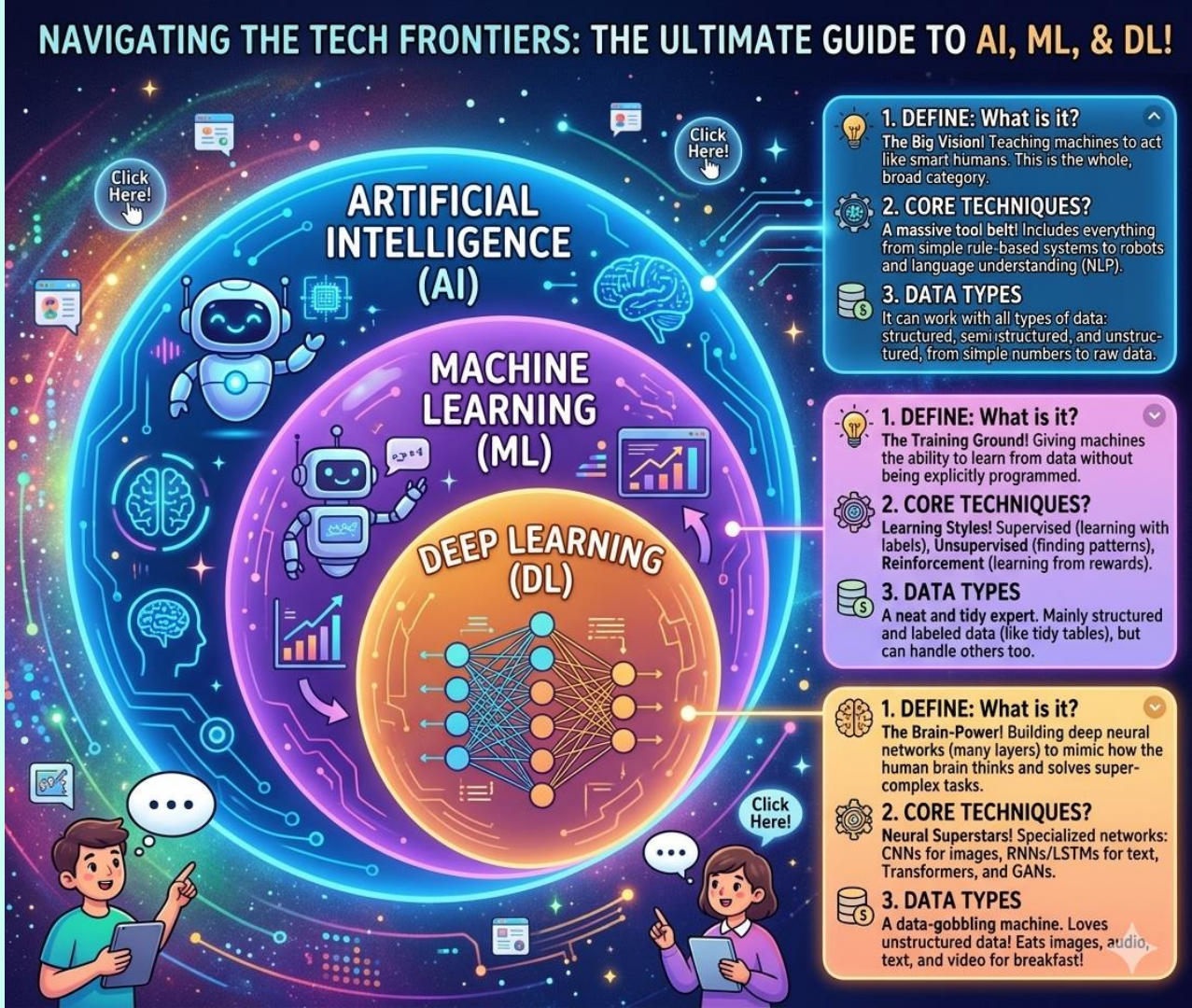
Introduction To Machine Learning

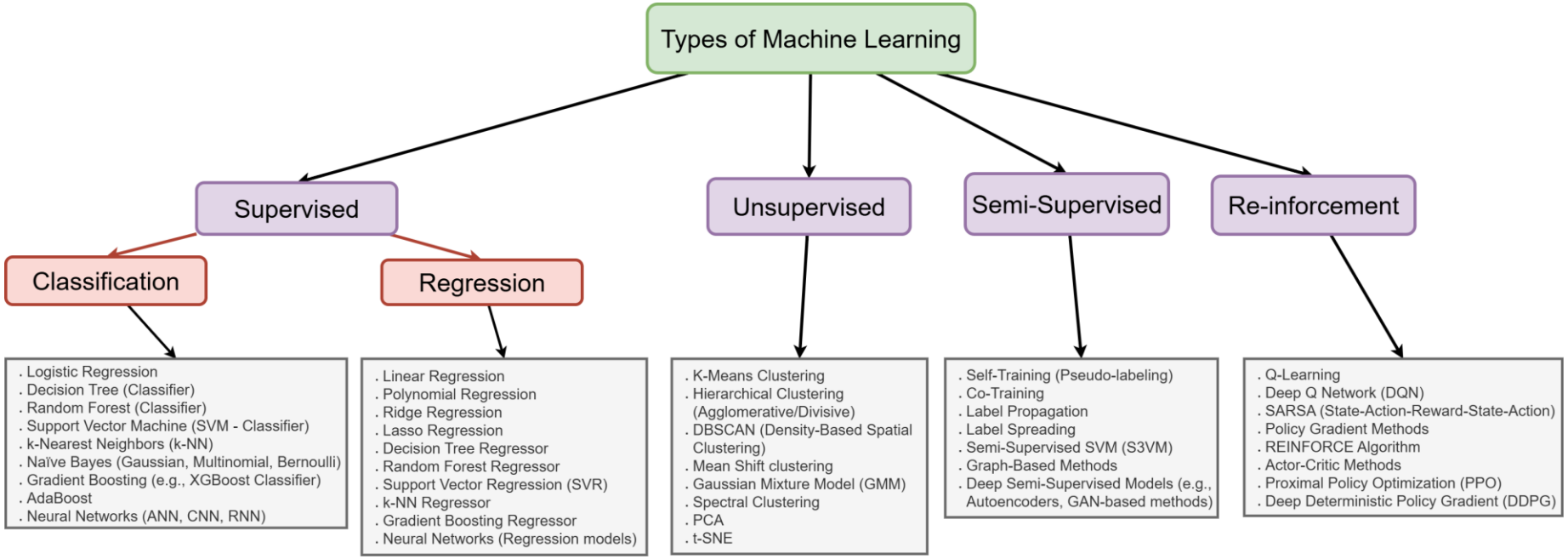
KJR

Content

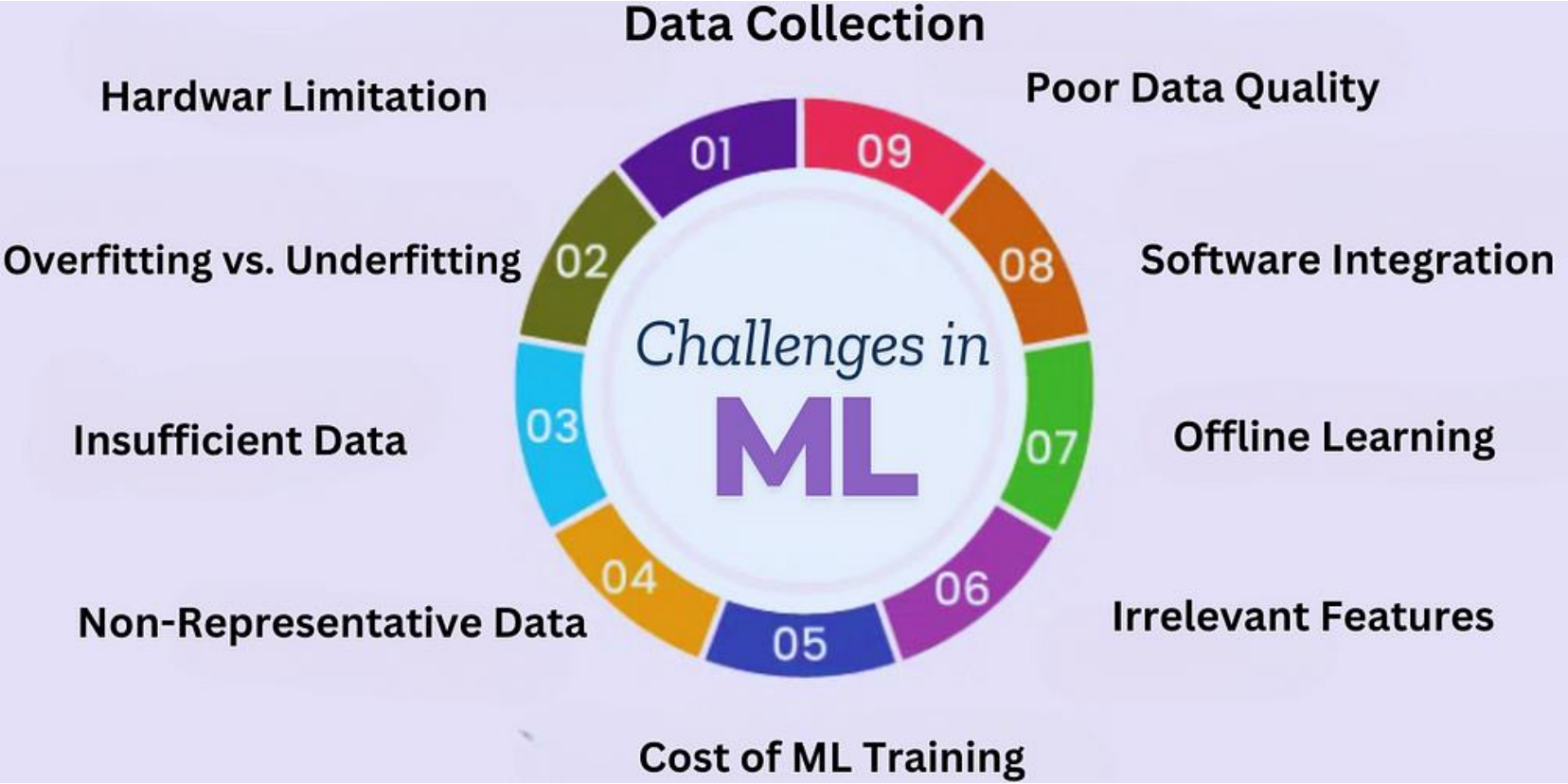
1. AI - ML - DL
2. Types of Machine Learning
3. Challenges of ML
4. Applications of ML
5. Machine Learning Lifecycle
6. Data Engineer VS Data Analyst VS Data Scientist VS ML Engineer

Difference Among
Artificial Intelligence (AI)
Machine Learning (ML)
Deep Learning (DL)



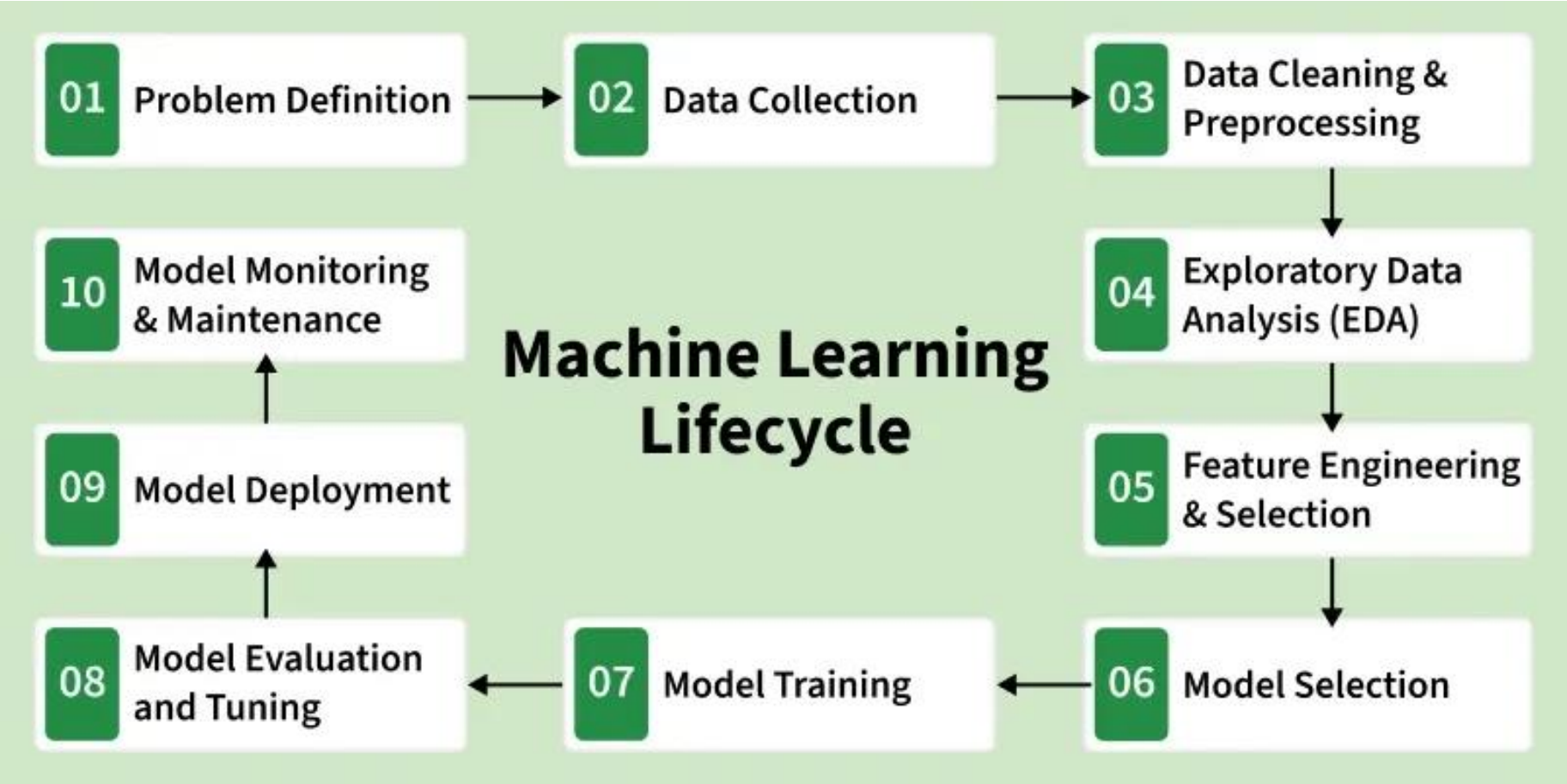

Types of Machine Learning
Supervised
Unsupervised
Semi-Supervised
Re-inforcement
Classification
Regression
. Logistic Regression
. Decision Tree (Classifier)
. Random Forest (Classifier)
. Support Vector Machine (SVM - Classifier)
. k-Nearest Neighbors (k-NN)
. Naïve Bayes (Gaussian, Multinomial, Bernoulli)
. Gradient Boosting (e.g., XGBoost Classifier)
. AdaBoost
. Neural Networks (ANN, CNN, RNN)
. Linear Regression
. Polynomial Regression
. Ridge Regression
. Lasso Regression
. Decision Tree Regressor
. Random Forest Regressor
. Support Vector Regression (SVR)
. k-NN Regressor
. Gradient Boosting Regressor
. Neural Networks (Regression models)
. K-Means Clustering
. Hierarchical Clustering (Agglomerative/Divisive)
. DBSCAN (Density-Based Spatial Clustering)
. Mean Shift clustering
. Gaussian Mixture Model (GMM)
. Spectral Clustering
. PCA
. t-SNE
. Self-Training (Pseudo-labeling)
. Co-Training
. Label Propagation
. Label Spreading
. Semi-Supervised SVM (S3VM)
. Graph-Based Methods
. Deep Semi-Supervised Models (e.g., Autoencoders, GAN-based methods)
. Q-Learning
. Deep Q Network (DQN)
. SARSA (State-Action-Reward-State-Action)
. Policy Gradient Methods
. REINFORCE Algorithm
. Actor-Critic Methods
. Proximal Policy Optimization (PPO)
. Deep Deterministic Policy Gradient (DDPG)

Challenges of ML



Applications of ML

Machine Learning Lifecycle



Different Job Roles

Data Engineer

1. Definition:

What does a Data Engineer do?

THEY BUILD THE PIPES!

They design the super-duper water slides for all our data.
They make sure data can flow easily from where it's kept to where the other smart folks can use it. They keep the data clean and organized!

2. Skills Required

- Being a builder of data slides!
- Working with big, complex sets of data.
- Good at organizing things.
- Understanding how databases work.

3. Tools:

SQL, Hadoop, Cloud Services

Data Analyst

1. Definition:

What does a Data Analyst do?

THEY TELL DATA STORIES!

They look closely at data to find cool secrets and answers.
For example: "Why do we sell more cookies on Mondays?"
They turn boring numbers into exciting picture reports to help make good decisions.

2. Skills:

- Spotting clues and patterns in data.
- Creating cool charts and graphs.
- Being good at simple math and logic.
- Communicating clearly so everyone understands the data story.

3. Tools:

SQL, X, +, ++++, ▥

Data Scientist

1. Definition:

What does a Data Scientist do?

THEY MAKE PREDICTIONS & RECIPES!

They build special number recipes, like spells, that can guess what will happen next. For example: "Which movie will you like the most?" or "How many ice creams will we sell?" They use fancy math to find the biggest insights.

2. Skills:

- Making smart number guesses with Machine Learning.
- Understanding big, tricky math (Statistics & Probability).
- Wrangling messy data into nice sets.
- Figuring out what info is most important (Feature Eng).

3. Tools:

Python, R, Neural Network

ML Engineer

1. Definition:

What does an ML Engineer do?

THEY MAKE MACHINES SMART!

They take the recipes and models from Data Scientists and make them work in the real world for millions of people. For example, the smart software that gives you movie recommendations. They ensure the AI is fast and reliable!

2. Skills:

- Creating smart 'Machine Learning' systems.
- Knowing how to code with Python or Java.
- Working with powerful computer brains (Big Data Tools like Spark).
- Building really fast, strong computer systems.

3. Tools:

TensorFlow, PyTorch, Spark like Hadoop

From Bangla book(PDF)

Basic Concept Discussion:

1. Learning types
2. Features
3. Train-test-validation
4. Cross validation

NEXT:

- Install Jupyter Notebook(anaconda/vscode/colab)
- Python basics required for ML

See You in the Next Lecture.....